Prob. 6. (Z & T, Ch 7. 8.2)

For the Forier series over the interval $[-\pi, \pi]$:

(1) $$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cdot \cos(nx) + b_n \cdot \sin(nx)\right],$$ show that

(2) $$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx$$

Soln: Let me multiply BS of (1) by $\sin(mx)$ & integrate from $[-\pi, \pi]$ interval:

(3) $$\int_{-\pi}^{\pi} f(x)\cdot\sin(mx)\,dx = \int_{-\pi}^{\pi}\left\{\frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n\cos(nx) + b_n\sin(nx)\right]\right\}\sin(mx)\,dx.$$

Now let me work on RHS of (3), using the linearity prop-y of $\int$:

$$RHS = \int_{-\pi}^{\pi}\left\{\frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n\cos(nx) + b_n\cdot\sin(nx)\right]\right\}\sin(mx)\,dx = \boxed{\int A + B\,dx = \int A\,dx + \int B\,dx}$$

$$= \int_{-\pi}^{\pi}\frac{a_0}{2}\sin(mx)\,dx + \underbrace{\int_{-\pi}^{\pi}\sum_{n=1}^{\infty}\left[a_n\cos(nx) + b_n\sin(nx)\right]\cdot\sin(mx)\,dx}_{I}.$$

I = can NOT do the linearity easily here, because we have infinite sum, rather than finite.

So, now RHS is

(4) $$RHS = \int_{-\pi}^{\pi}\frac{a_0}{2}\sin(mx)\,dx + I,$$ where

(5) $$I = \int_{-\pi}^{\pi}\sum_{n=1}^{\infty}\left[a_n\cos(nx) + b_n\cdot\sin(nx)\right]\cdot\sin(mx)\,dx$$

To change $\int_{-\pi}^{\pi}$ & $\sum_{n=1}^{\infty}$ in (5), now let me assume some properties

that allows this interchange:

Assume

(6) $$\sum_{n=1}^{\infty} \left[a_n \cos(nx) + b_n \cdot \sin(nx)\right] \cdot \sin mx$$ is a <u>uniform-convergent</u> series. ! This very <u>important</u> assumption to go further in (5).

For uniform convergence of (6) we can use (require to have a_n, b_n's s.t. satisfy) weierstrass - M test:

Thm (Weierstrass - M test).

Let M_n be a sequence of <u>positive</u> real numbers s.t. for

(7) $$g_n(x) := \left[a_n \cos(nx) + b_n \cdot \sin(nx)\right] \sin(mx),$$

(8) $$|g_n(x)| \leq M_n$$

for $\forall x \in [-\pi, \pi]$, $n \in \mathbb{N}$. If the series $\sum M_n$ is <u>convergent</u> then $\sum g_n$ (the series in (6)) is <u>uniformly</u> convergent on $[-\pi, \pi]$.

{ Side note:

To satisfy (8), we must have

(9) $$\left|\left[a_n \cos(nx) + b_n \cdot \sin(nx)\right] \sin(mx)\right| \leq |a_n + b_n| \overset{\Delta \text{ ineq.}}{\leq} |a_n| + |b_n|.$$

$\sin, \cos \leq 1$

- Hence the coef-ts a_n, b_n <u>must satisfy</u>

(10) $$|a_n| + |b_n| \leq M_n$$ for some M_n with the property $\sum M_n$ <u>converges</u>.

Now with established property of (6), being uniformly convergent, let me continue (5) as follows.

because of unif. conv.

$$(11') \quad I = \int_{-\pi}^{\pi} \sum_{n=1}^{\infty} \left[a_n \cos(nx) + b_n \sin(nx) \right] \cdot \sin(mx)\,dx =$$

$$= \sum_{n=1}^{\infty} \left\{ \underbrace{\int_{-\pi}^{\pi} a_n \cos(nx) \sin(mx)\,dx}_{=0,\text{ always, by result of Pr. 5(c)}} + \underbrace{\int_{-\pi}^{\pi} b_n \sin(nx) \cdot \sin(mx)\,dx}_{=0 \text{ unless } n=m\text{, by Pr. 5 (e)}} \right\} =$$

const.

$$= \left\{ 0 + b_m \int_{-\pi}^{\pi} \sin(mx) \cdot \sin(mx)\,dx \right\} = b_m \underbrace{\int_{-\pi}^{\pi} \sin^2(mx)\,dx}_{=\pi \text{ by Pr. 5, (8)}}$$

$$= b_m \cdot \pi. \quad \Rightarrow$$

$$(11) \quad I = \int_{-\pi}^{\pi} \sum_{n=1}^{\infty} \left[a_n \cos(nx) + b_n \sin(nx) \right] \cdot \sin(mx)\,dx = b_m \cdot \pi.$$

Then, the place we picked off, RHS is as follows:

$$(12) \quad RHS = \underbrace{\frac{a_0}{2} \int_{-\pi}^{\pi} \sin(mx)\,dx}_{=0,\text{ by Pr. 5, (b)}} + I = 0 + I = b_m \cdot \pi.$$

Then putting this (12) to the RHS of (3), we get

$$\int_{-\pi}^{\pi} f(x) \sin(mx)\,dx = b_m \cdot \pi \quad \Rightarrow$$

$$b_m = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cdot \sin(mx)\,dx,$$ as desired, provided the "nice" conditions met in (7)–(10). ∎